MAHENDRA PAL BHALOTHIYA

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in **data analysis**, **predictive modeling**, and building scalable **generative AI pipelines**. Proficient in **Python**, **SQL**, and **REST APIs** to identify business needs, extract and validate multi-source data, and perform quantitative analyses. Adept at creating financial and operational models in **Excel / Google Sheets** and communicating actionable recommendations to stakeholders.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, SQL, R, Pandas, NumPy, Data Extraction, Data Validation, Scikit-learn
- **Financial & Spreadsheet Modeling:** Advanced Excel & Google Sheets Modeling, Quantitative Analysis, Statistical Modeling
- **Generative AI & Machine Learning:** LLMs, OpenAI APIs, Hugging Face, LangChain, PyTorch, TensorFlow, Predictive Modeling
- **Software Development & Web:** React.js, Node.js, REST APIs, FastAPIs, Docker, AWS (S3, Lambda, EC2), Git / GitHub
- **Core Competencies:** Data Structures & Algorithms, Database Design, System Architecture, Technical Documentation

EXPERIENCE

AI/ML Research Intern **May 2025 – Jun 2025**

IIT Guwahati (AI/ML Research Group) *Guwahati, India*

- Scoped an end-to-end computer vision business problem, designing data extraction, preparation, and validation pipelines for live video feeds.
- Developed and validated an object detection and alerting model in Python, improving detection accuracy by 18% and translating complex technical outputs into actionable insights for non-technical stakeholders.
- Analyzed quantitative model performance metrics using SQL and Pandas to identify failure modes, presenting findings to senior modelers to refine system architecture and pipeline stability.

RELEVANT PROJECTS

Startup Ecosystem Portal – Multi-Stakeholder Data Platform *Python, SQL, PyTorch, Scikit-learn, REST APIs*

- Engineered a full-stack data platform with custom workflow systems, data dictionaries, and dashboard analytics for entrepreneurs, investors, and customers.
- Integrated a PyTorch-based conversational LLM and a Random Forest funding-prediction model; designed REST APIs to extract and validate data seamlessly across system modules.

Market Sentiment & Quantitative Crypto Analysis *Python, SQL, Pandas, Matplotlib, Predictive Analytics*

- Analyzed multi-year trading datasets alongside the Crypto Fear & Greed Index using statistical modeling to identify correlation patterns and behavioral trends.
- Built predictive analytics scripts and data visualization dashboards to surface actionable, data-driven trading strategies and key performance metrics.

AI-Diagnostic – Adaptive Testing & Learning Platform *Python, Applied NLP, Item Response Theory, LLM APIs*

- Developed an adaptive testing engine incorporating Item Response Theory (IRT) and LLM-driven analytics for real-time ability estimation.
- Implemented real-time data collection and LLM API integrations to generate personalized feedback reports and actionable study recommendations.

EDUCATION

National Institute of Technology Mizoram **Aug 2023 – Present**

Bachelor of Technology in Computer Science & Engineering *CGPA: 7.60 / 10.0*

Army Public School, Jaipur **2020 – 2022**

Intermediate (91.8%) • Matriculation (89.2%) • JEE Mains: 96.97 Percentile